Roll No.: ....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: **B. Tech**

Semester: III

Name of the Course: Networks Analysis and Synthesis

Course Code: TEC 303

Time: 90 Minutes

Maximum Marks: 50

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1	10 Marks	CO1
(a)	Use mesh analysis find the loop currents of Fig. 1.	

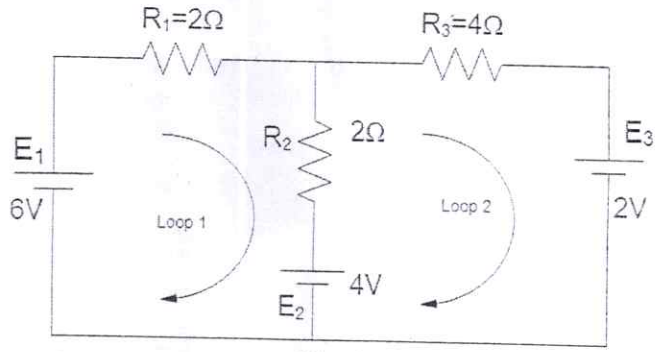

Fig. 1

	OR	
(b)	Find the power for the 6V source and indicate whether the source is delivering or absorbing power, from Fig. 1 above.	
Q2	10 Marks	CO1
(a)	A circuit consists of a battery of 18 V and three resistors: R1=3 Ω, R2=6 Ω, and R3=9 Ω. Resistors R1 and R2 are connected in parallel, and their combination is connected in series with R3. Calculate the equivalent resistance of the circuit and current in each resistor (R1, R2, and R3)	CO2
	OR	
(b)	Find the current through 6 Ω resistor using superposition theorem in Fig. 2.	

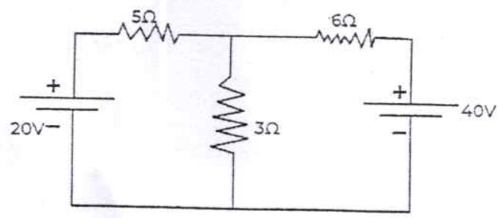

Fig. 2

Q3 (a)	Find the Thevenin's equivalent circuit (V_{th} and R_{th}) across ab terminal in Fig.3.	10 Marks	CO1 CO2

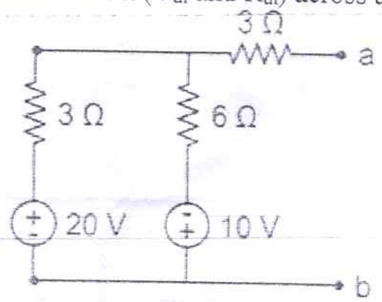

Fig.3

OR

(b) Define the following terms,

(I) Link (II) Graph (III) Tree (IV) Node (V) Branch (VI) Twig of the following network and find incidence matrix.

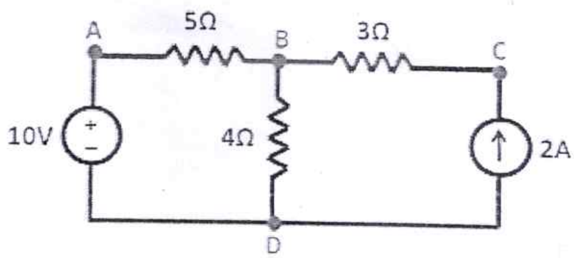

Fig.4

Q4 (a)	Draw the oriented graph of the network with fundamental cut set matrix as shown	10 Marks	CO2

	Branchase ⇒	1	2	3	4	5	6
Cutsets							
C_2		+1	-1	0	0	-1	0
C_3		0	0	-1	0	+1	-1
C_4		-1	0	0	+1	0	+1

OR

(b) Explain maximum power transfer theorem and reciprocity theorem with a proper example?

Q5 (a)	Explain Superposition Theorem and its limitations. Solve an example to illustrate its use.	10 Marks	CO1 CO3

OR

(b) In the circuit shown below, the switch "s" is open at t= 0, with zero initial conditions, find v, dv/dt, d^2v/dt^2 at $t = 0^+$

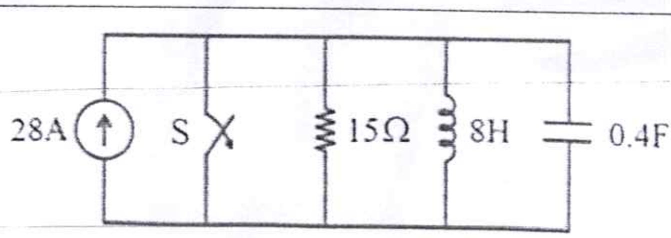

Fig. 8